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Your grade: 100%

Your best: 100% - Your highest score

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Advanced Markov Chain Methods for Bayesian Computation Quiz

Instructions

Review Learning Objectives

1. Based on the reading, what is an ideal property of a Metropolis-Hastings jumping distribution?

1 / 1 point

- ☐ It should be symmetric. **Correct**
- ☐ It should always lead to **rejection** when you've learned before starting the assignment? I'm here to help.
- ☒ It should have a high **acceptance rate**. **Correct** Let's chat
- ☐ It should be difficult to sample from.

☒ Correct

Correct! A good jumping distribution should not be rejected too frequently, ensuring a reasonable acceptance rate.

Submitted

Dec 5, 2:18 AM IST

Attempts

Unlimited

Retry

2. What is the Gibbs sampler particularly useful for?

1 / 1 point

- ☐ Deterministic integrations you need at least 80% We keep your highest score.
- ☐ Models where conditional distributions are difficult to sample from.
- ☒ Multi-dimensional problems with conditionally conjugate models. **Correct**
- ☐ One-dimensional models.

☒ Correct

Correct! The Gibbs sampler is particularly useful for multi-dimensional problems where the conditional distributions are conjugate.

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3. What is the Metropolis algorithm based on?

1 / 1 point

- ☐ A fixed step size.
- ☐ Deterministic sampling.
- ☐ Sampling directly from the posterior distribution.
- ☒ A random walk with an acceptance/rejection rule. **Correct**

☒ Correct

Correct! The Metropolis algorithm is based on a random walk with an acceptance/rejection rule to converge to the target distribution.